

{DALAS} Drug Repurposing Project

Paul Béglin - 21408798

Vladislav Antipin - 21242244

Predicting new drug-disease associations helps to narrow down the list of potential candidate molecules allowing researchers to set priorities for new clinical trials. The goal of our project is to use public chemical and medical datasets to predict those associations.

The key data source for this project is ChEMBL - a huge SQL-based relational database of bioactive molecules, bringing together chemical, bioactivity, and genomic data. Fortunately, it provides a Python API client that allows efficient querying and returns JSON responses. We will complement the information from this database with other public biomedical databases described below.

We focus on autoimmune diseases because they share common biological mechanisms (e.g. inflammation pathways, immune dysregulation, cytokine signaling). This increases the likelihood of drug repurposing, as treatments effective for one autoimmune disease (e.g. rheumatoid arthritis) may also benefit others (e.g. lupus or Crohn's). Moreover, autoimmune diseases are clinically well-documented and rich in public data, which facilitates model validation. In order to extract their MeSH IDs, we've scraped them from the NIH site listing the disease classification and retrieved 50 identifiers. Then, we've used them to get known drug indications (1,256) for these diseases using ChEMBL API and used the same service to get chemical features and basic information on those drugs (515) - biological (125) and small molecules (390).

Our prediction target is an indicator of drug-disease association (approved medical indication) retrieved from the ChEMBL database. It can be further organized either as a binary label for each drug-disease pair for simple baseline classification approaches such as Logistic Regression, Random Forest, and XGBoost, or as incidence matrix for more advanced graph-based models. The interest of Graph Neural Networks follows from the fact that most of this data has intrinsic graph structure - drugs and diseases can represent nodes of bipartite graph, each having its feature payload, edges represent established indications. Furthermore, some features such as molecular targets of drugs and molecular actors of diseases represent nodes bridging two layers of this bipartite graph.

Considering all combinations of 515 drugs with 50 diseases (25,750 pairs), we have only around 5% of combinations (1,256) that are actually approved indications, so the classes are imbalanced as we

have expected. It means that we would have to use balancing techniques for model training and evaluation.

Data completeness and consistency vary across databases. In some cases the missing data is natural and inevitable, e.g. presence of some chemical features depend on the drug type (biological or small molecule), and we will need to either model them separately with models requiring homogeneous feature data (e.g. Linear Regression) or use more flexible graph models. In other cases, we will assess missingness rates and apply imputation where feasible (e.g. mean or KNN imputation for numerical features, “unknown” categories for nominal features). For more complex features (e.g. embeddings), we might drop entities with insufficient metadata to maintain data integrity.

We envisage enriching of chemical features with tools such as RDKit, e.g. retrieving numeric representation of chemical structures. Using the same ChEMBL API, we can retrieve the data about molecular targets of those drugs and complement them with target information from the DrugBank dataset (plain csv files) by matching drug identifiers. Furthermore, drug indications from ChEMBL provide references to corresponding clinical trials of ClinicalTrials.gov, we plan to leverage this to get the information such as timestamps of approval using their API. For disease features such as molecules / genes involved in pathogenesis, we will address DisGeNET and MeSH APIs. The tricky part might be the matching of identifiers of drug targets and molecules with corresponding genes involved in diseases. We plan to also extract the text embeddings from the descriptions of drugs and diseases using BioBERT and PubMedBERT and use them to compute the proximity index. Another approach might be to compute a scientific co-citation index for each drug-disease pair by parsing PubMed abstracts.

One of potential weaknesses of our project is that at the moment we consider only approved drug-disease indications, but don't distinguish drugs with not (yet) proven activity from those that have been proved to be inefficient. We plan to fix it by analyzing clinical trials metadata from ClinicalTrials.gov though it is uncertain whether this will succeed. As for potential biases, text embeddings (from PubMedBERT) might reflect research frequency, not true biological relationships.

Considering legal aspects, all datasets we use are public and contain no personal information, The only exception is that we requested academic access to the DrugBank database. Speaking of scraping policies, biomedical sites discourage scraping and insist on usage of their APIs, so we scraped only in cases where it was more straightforward to parse the webpage (e.g. get autoimmune diseases ids and probably we'll have to use it to get some clinical trials metadata or textual descriptions of diseases for LLMs).